

Harshith Pabbati

Software Engineer

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Summary

Software engineer with 4+ years of experience building real-time communication and performance-focused software. Worked on **WebRTC and SCTP infrastructure at Cloudflare**, live streaming and developer tooling at Daily.co, and web performance and analytics at Aragon.ai. Currently pursuing an **MS in Computer Science at Stony Brook University**, after completing a Summer 2026 internship with Cloudflare's Realtime team.

Education

Stony Brook University

Master of Science in Computer Science

Aug 2025 – May 2027

Stony Brook, NY

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science and Engineering

Aug 2018 – May 2022

Kerala, India

Experience

Cloudflare

Software Engineer Intern, Realtime

May 2026 – Aug 2026

Austin, TX

- Replaced a **TCP**-based inter-SFU relay hop with native **SCTP routing**, supporting reliable and unreliable, ordered and unordered delivery while eliminating head-of-line blocking for latency-sensitive traffic.
- Improved median throughput by **29%**, reduced jitter by **17.6%**, and lowered p99 RTT by **11.3%** by engineering a **fragment passthrough** path through pion/sctp, allowing large DataChannel messages to skip full reassembly at intermediate SFUs.
- Designed and shipped **bidirectional DataChannels (canReply)**, allowing subscribers to reply to publishers over the same channel and enabling use cases such as AI agents and remote control.

Aragon.ai

Software Engineer

Jun 2024 – Jul 2025

Remote

- Improved **performance by 50%** across website, reducing LCP from **20s to 2.1s**, FCP from **4s to 0.4s**, and CLS from **0.20 to 0.03** through code splitting, lazy loading, and caching.
- Migrated product analytics from **Mixpanel to PostHog** to reduce costs, moving existing data, rebuilding event tracking, and using funnel analytics to better understand conversion through key user flows.

Daily.co

Software Engineer

Nov 2021 – May 2024

Remote

- Delivered Daily's first comprehensive developer-facing showcase for **Interactive Live Streaming APIs** using React and Next.js, with reusable applications that helped customers evaluate and launch real-time products.
- Developed an internal meeting intelligence platform using Daily APIs with **AI summaries, sentiment analysis, and talk-time metrics**, adopted by customer-facing teams across **1,500+ meetings**.

Projects

Daily Studio | Browser-based Live Streaming Studio

- Built a browser-based live streaming studio supporting **100K+ concurrent viewers at ~200ms latency**, with real-time bring-to-stage interactions, live chat, viewer analytics, and multi-platform streaming.
- Implemented a customizable **drag-and-drop video composition system** using Daily's Video Component System (VCS) and Interactive Live Streaming (ILS) APIs in Next.js.

Technical Skills

- **Languages:** TypeScript, JavaScript, Go, Rust, Python, SQL
- **Systems & Networks:** WebRTC, SCTP, DataChannels, SFU, Distributed Systems, Performance Optimization
- **Backend & Data:** Node.js, Express.js, Django, REST APIs, GraphQL, PostgreSQL, gRPC
- **Frontend:** React.js, React Native, Next.js, HTML5, CSS3
- **Tools:** Git, Gerrit, Github, GitLab, VS Code, Postman, LLM-assisted development
- **Open Source:** GraphQL Foundation (Linux Foundation Mentorship); GSoC Mentor: Public Lab